

1. What is a Database?

A database is an organized collection of data that is stored electronically so that it can be easily accessed, managed and updated. Databases are essential because they allow businesses and organizations to store large amounts of information securely and retrieve it quickly whenever needed.

Examples:

1. A bank uses databases to store customer account information and transaction records.
2. A hospital uses databases to store patient records, medical histories and appointment schedules.

Source: IBM Database Fundamentals, Microsoft Learn.

2. What is a Data Warehouse?

A data warehouse is a centralized system that stores historical data from different sources for reporting and analysis purposes.

Difference from a regular databases:

- A database is designed for day to day operations.
- A data warehouse is designed for analyzing large amounts of historical data.

Why companies use data warehouses: Companies use data warehouses to identify trends, create reports and support business decision making.

Example: A retail company may analyze several years of sales to identify purchasing trends.

3. What is a Data Lake?

A data lake is a storage system that holds large amounts of raw data in its original format.

Difference from data warehouse:

- Data warehouse store structured and processed data.
- Data lakes store structured, semi-structured and unstructured data.

When businesses use a data lake: A company might use a data lake when collecting data from mobile apps, sensors, videos, social media and websites that may be analyzed later.

4. What is a Data Mart?

A data mart is a smaller section of a data warehouse that is designed for a specific department or business unit.

Relationship to a data warehouse: A data mart usually obtains its data from a larger data warehouse.

Typical users:

- Sales teams
- Finance departments
- Human Resources departments

Example: The finance department may have a data mart containing only financial information.

5. Structured, Semi-Structured and Unstructured Data

structured Data

- Data organized into rows and columns.

Example: Customer information stored in a Excel spreadsheet.

Semi-Structured Data

Data that does not fit neatly into tables but still contains labels or tags.

Example: JSON files.

Unstructured Data

Data with no predefined format.

Example: Video, photographs, audio files and Social Media posts

6. What is Data Modeling?

Data modeling is the process of designing how data will be stored, connected and organized within a System.

Importance:

- Improves data quality
- Reduces duplication
- Makes systems easier to maintain.
- Improves performance

Types of Data Model

Conceptual Data Model - provides a high level view of business requirements.

Logical Data Model - Defines relationships, data elements and rules without considering a specific database system.

Physical Data Model - Shows exactly how data will be stored in a database.

7. What is Data Mining?

Data mining is the process of analyzing large datasets to discover patterns, relationships, trends and useful information.

Business uses:

- Predicts customer behavior
- Detects fraud
- Improve marketing campaigns
- Increase sales

Examples

1. Banks use data mining to detect suspicious transactions and prevent fraud.
2. Online retailers use data mining to recommend products customers may want to buy

8. What is a DBMS?

A database Management system (DBMS) is a software used to create, store, manage and retrieve data from databases.

MySQL - a popular open-source relational database.

Use Case: Websites and online applications

PostgreSQL - An advanced open-source relational database.

Use Case: Business applications requiring complex data analysis.

Oracle Database - A commercial database system designed for large enterprises.

Use Case: Banking and large corporate systems.

MongoDB - A NoSQL database that stores data in document format.

Use Case: Applications handling large volumes of flexible data such as social media platforms.

9. What is ETL?

ETL stands for Extract, Transform, Load.

Extract - Data is collected from different sources such as databases, spreadsheets and websites.

Transform - Data is cleaned, corrected and converted into a consistent format.

Load - The cleaned data is loaded into a system such as a data warehouse.

Importance of ETL: ETL ensures that data is accurate, consistent and ready for analysis.

10. Common Data Storage Formats

CSV (comma-separated values)

Stores data in rows and columns separated by commas.

Used for: APIs and web applications

Parquet - A column-based storage format optimized for analytics.

Used for: Big data processing and data warehouses.

Avro - A compact binary storage format with built-in schemas.

Used for: Data streaming and large-scale data processing

11. What is Artificial Intelligence (AI)?

Artificial Intelligence (AI) is a branch of computer science that focuses on creating machines and software that can perform tasks that normally require human intelligence. These tasks include learning, problem-solving, decision-making, understanding

language and recognizing images.

Narrow AI (Weak AI)

- Narrow AI is designed to perform a specific task.

Example: Siri, Google Assistant or Netflix recommendations.

General AI (strong AI)

General AI refers to a theoretical type of AI that can think, learn and perform any intellectual task that a human can do.

Example: General AI does not yet exist, but it is often shown in science fiction movies such as JARVIS from Iron Man.

12. What is Machine Learning (ML)

Machine Learning is a subset of AI that enables computers to learn from data and improve their performance without being explicitly programmed for every task.

Differences from Traditional Programming

Traditional Programming

- Programmer writes rules
- Data and rules produce answers

Machine Learning

- Data and answers are used to create rules automatically.

Types of Machine Learning

Supervised Learning - The model learns using labeled data.

Example: Predicting whether an email is spam or not spam.

Unsupervised Learning - The model discovers patterns in data without labels.

Example: Grouping customers with similar buying habits.

Reinforcement Learning - The model learns through rewards and penalties.

Examples: Self-driving cars and game-playing AI.

13. What is Deep Learning?

Deep Learning is a specialized area of Machine Learning that uses artificial neural networks with many layers to analyze large amounts of data.

Relationship Between AI, ML and Deep Learning

Artificial Intelligence



Machine Learning



Deep Learning

What are Neural Networks?

Neural networks are computer inspired by the human brain. They consist of connected nodes called neurons that process information.

How They Work

- Input data enters the network
- Hidden layers analyze patterns
- Output layer produces a result.

Example: Identifying whether a picture contains a dog or a cat.

14. What is Natural Language Processing (NLP)?

Natural language processing (NLP) is a branch of AI that helps computers understand, interpret and generate human language.

Everyday Applications of NLP

- Voice assistants such as Siri and Google assistant.
- Language translation tools such as Google translate.
- Predictive text and autocorrect on smartphones.

These systems analyze text and speech to understand meaning and provide useful responses.

15. What is Large Language Model (LLM)

A large language model (LLM) is an advanced AI model trained on massive amounts of text data to understand and generate human-like language.

Difference Between an LLM and Standard ML Model

Large Language Model

- Can perform many language tasks
- Uses extremely large datasets
- Highly adaptable

Standard ML Model

- Designed for specific tasks
- Uses smaller datasets
- Limited flexibility

Examples of LLMs

ChatGPT - Used for answering questions, writing content, summarizing information, coding assistance and learning

16. What is Generative AI?

Generative AI is a type of AI that creates new content such as text, images, audio, videos or code.

How it works (High Level)

- The AI learns patterns from large datasets.
- It identifies relationships within the data.
- It generates new content based on those patterns.

Examples of Generative AI Tool

- Chat GPT - Generates text, explanations, summaries and code
- Microsoft Copilot - Assists with writing, research, coding and productivity tasks.
- DALL-E - Creates images from written descriptions.

17. What is Training Dataset?

A training dataset is a collection of data used to teach an AI or Machine Learning model how to perform a task.

Why Data Quality Matters

High-quality data helps a model:

- Learn accurately
- Make reliable predictions
- Reduce errors

Problems Caused by Poor Data

If training data is biased or incorrect:

- The AI may make unfair decisions
- Predictions may be inaccurate.
- Certain groups of people may be treated unfairly.
- The model may not perform well in real-world situations.

Example: If a hiring AI is trained mostly on data from one group of people, it may unfairly favor similar candidates.

18. What is Computer Vision?

Computer Vision is a field of AI that enables computers to interpret and understand images and videos.

How Machines "see"

1. A camera captures an image.
2. The image is ~~code~~ converted into digital data.
3. AI algorithms analyze shapes, colours, edges and patterns.

4. The system identifies objects and provides results.

Real-World Uses of Computer Vision

Facial Recognition - used in smartphones to unlock devices securely.

Medical Imaging - Used by doctors to help detect diseases in X-rays, CT scans and MRI images.

19. What is an API (Application Programming Interface)?

An API is a set of rules that allows different software applications to communicate and share information with each other. It acts as a bridge between systems.

Real-World Analogy

Think of an API as a waiter in a restaurant:

- You (The customer) place an order
- The waiter takes the order to the kitchen
- The kitchen prepares the food.
- The waiter brings the food back to you.

The waiter is like an API because it allows communication between you and the kitchen without either side needing to know all the details of how the other works.

Why APIs are Important

- Enable systems to communicate
- Allow data sharing between applications.
- Save development time
- Support integration between different services.

Example: A weather app uses an API to get weather information from a weather service.

Q. What is an API key?

An API key is a unique code used to identify and authenticate a user or application when accessing an API.

Purpose of an API key

- Controls access to the API.
- Tracks usage.
- Prevents unauthorized access.

How It Should Be protected

- Never share it publicly
- Store it securely
- Do not place it in public code repositories

Risks of Exposure

If an API key becomes public:

- Unauthorized users may access the API.
- Extra costs may be charged to the owner.
- Sensitive data may be exposed.
- The API provider may ~~be~~ suspended the account.

GET

- Retrieves information

Example: Getting customer details from a database.

POST

- Creates new information

Example: Creating a new customer account.

PUT

updates existing information.

Example: updating a customer's email address.

DELETE

- Removes information

Example: Deleting an account record.

Q2. What is JSON (JavaScript Object Notation)?

JSON is a lightweight data format used to store and exchange information between systems.

Why JSON is popular

- Easy for humans to read.
- Easy for computers to process.
- Widely supported by programming languages.
- Commonly used with APIs.

Q3. What is prompt Engineering?

Prompt Engineering is the process of writing clear and effective instructions of AI systems so that they produce better results.

23 Why It is Important

- Improves AI responses
- Saves time.
- Produces more accurate results.
- Helps users achieve their goals efficiently.

Example 1

Poor Prompt:

Tell me about cars.

Improved Prompt:

Explain the main features, fuel economy and safety systems of the 2024 Toyota Hilux in 200 words.

Example 2

Poor Prompt:

Write an email.

Improved Prompt:

Write a professional email to a client thanking them for their business and confirming a meeting on Friday at 10:00 AM.

24. What is a MAP Function?

A MAP Function is used in programming to apply the same operation to every item in a list and return a new list containing the results.

Python Example

```
numbers = [1, 2, 3, 4]
```

```
result = list(mapmap(lambda x: x*2, numbers))
```

```
Print(result)
```

Output

```
[2, 4, 6, 8]
```

Why It is Useful

- Reduces repetitive code.
- Processes data efficiently

- Makes programs easier to read.

25. What is Cloud Computing?

Cloud Computing is the delivery of computing services such as storage, networking, databases and software over the internet.

instead of buying and maintaining physical servers, companies can rent resources from cloud providers.

Main Cloud Service Models

IaaS (Infrastructure as a service) - provide virtual servers, storage and networks.

Example: Amazon EC2 (AWS)

PaaS (Platform as a service) - provides a platform for developing and deploying applications.

Example: Microsoft Azure App Service

SaaS (Software as a Service) - provides software that users access through the internet.

Example: Microsoft 365

26. What is Version Control system?

A version control system (VCS) is a software that tracks changes made to files and code over time.

It allows teams to collaborate and provides a history of modifications.

What is Git?

Git is the world's most widely used version control system. It helps developers and analysts

manage projects and work together efficiently.

Why Git is Popular

- Tracks changes
- Supports teamwork

- allows rollback to earlier versions
- Maintains a history of updates.

What is a Commit?

A commit is a saved snapshot of work at a specific point in time.

Example: Saving changes after adding a new feature.

What is Branch?

A branch is a separate version of a project where changes can be developed without affecting the main version.

Example: Creating a branch to test a new feature before merging it into the main project.

27. What is Jupyter Notebook?

A Jupyter Notebook is an interactive tool used to write and run code, display results, create visualizations and explain findings in one document.

Most Common Programming Language

Python is the language most commonly used in Jupyter Notebooks.

Why It Is Popular

- Combines code and explanations.
- Supports charts and visualizations.
- Makes data analysis easier.
- Ideal for experimentation and learning.

Common Uses

- Data analysis
- Machine learning
- Data visualization
- Research projects

28. What is Python?

Python is high-level programming language known for its simplicity and readability.

It is one of the most popular languages for Data Analytics, Data Science and Artificial Intelligence.

Why Python is Popular

- Easy to learn
- Large Community support
- Extensive libraries
- Works well with data and AI projects.

Common Python Libraries

Pandas - used for data manipulation and analysis.

Example: Cleaning and analyzing spreadsheets.

Numpy - used for mathematical and numerical operations.

Example: Performing calculations on large datasets

Matplotlib - used for creating charts and paragraphs.

Example: Building sales trend reports.

Scikit-Learn - used for machine learning.

Example: Creating models that predict customer behaviour.

What is Data Privacy?

Data Privacy refers to the protection of personal and sensitive information and ensuring that it is collected, stored, used and shared responsibly.

Why Data Privacy is Important

- Protects personal information.
- Prevents identity theft.
- Builds trust between organization and customers.
- Ensures compliance with laws and regulations.

What is GDPR?

The General Data Protection Regulation (GDPR) is data privacy law introduced by the European Union (EU) in 2018. It gives individuals more control over how their personal information is collected and used.

Countries where GDPR Applies

- All European (EU) member countries.
- European Economic Area (EEA) countries.
- Any organization worldwide that processes the personal data of EU citizens.

Main Principles of GDPR

- Data must be collected lawfully.
- Individuals must give consent.
- Personal data must be protected.
- Organizations must report serious data breaches.

30. What is Data Ethics?

Data Ethics refers to the moral principles and standards that guide how data is collected, used, shared and stored.

It ensures that organizations use data fairly, responsibly and transparently.

Example 1: Using Personal Data without Consent

- A company collects customer information and sells it to advertisers without permission.

Why it is unethical: It violates privacy rights and trust.

Example 2: Biased Hiring Algorithms

An AI system favors applicants from certain backgrounds while unfairly rejecting others.

Why it is unethical: It can result in discrimination and unfair treatment.

31. What is Data Governance?

Data governance is the framework of policies, processes, standards and responsibilities used to manage an organization's data.

It ensures that data is accurate, secure, accessible and used appropriately.

Key Components of a Good Data Governance framework

Data Quality - Ensures data is accurate, complete and reliable.

Data Security - Protects data from unauthorized access.

Data Privacy - Ensures compliance with privacy regulations.

Data Ownership - Defines who is responsible for managing data.

Data Standards - Establishes consistent rules for collecting and storing data.

Benefits of Data Governance

- Better decision-making
- Improved compliance
- Reduced risk.
- Greater trust in information.

32. What is Bias in AI?

Bias in AI occurs when an AI system produces unfair or inaccurate results because of problems in the data used to train it or the way it was designed.

Examples of AI Bias

A facial recognition system performs accurately for some groups of people but less accurately for others because the training data contained more examples from one group than another.

Potential Consequences

- Unfair decisions
- Discrimination
- Loss of trust
- Legal and ethical issues
- Reduced accuracy

How Bias can be Reduced

- Use ~~div~~ diverse data
- Test systems regularly
- Monitor results.
- Include different perspectives during development.

33. Differences Between Data Analyst, Data Scientist, Data Engineers and AI engineer

Role	Main Responsibility	Key Skills	Common Tools
Data analyst	Analyze data and create reports	SQL, Excel, Power BI	Excel, Power BI, SQL
Data Scientist	Build predictive models and find insights	Statistic, Machine Learning, Python	Python, R, Jupyter
Data Engineer	Build and maintain data pipelines	Databases, ETL, Cloud Platforms	Spark, SQL, Azure
AI Engineer	Design and deploy AI systems	Machine Learning, Deep Learning	TensorFlow, PyTorch, Python

Simple Explanation

Data Analyst: Answers "what happened"

Data Scientist: Answers "why did it happen" and "what may happen next"

Data Engineer: Builds the systems that move and store data.

AI Engineer : Builds AI application and machine learning solutions.

34. What is Business Intelligence (BI)?

Business Intelligence (BI) is the process of collecting, analyzing and visualizing data to support better business decisions.

BI helps organizations convert raw data into useful insights.

Popular BI Tools

Power BI - Power BI is Microsoft's business intelligence platform.

How it helps businesses:

- Creates interactive dashboards.
- Tracks business performance.
- Support data-driven decisions.

Tableau - Tableau is a data visualization and analytics tool.

How it helps businesses:

- Creates visual reports.
- Identifies trends.
- Simplifies complex data.

Benefits of BI

- Improves decision-making.
- Identifies trends and opportunities.
- Increases efficiency.
- Enhances business performance.

35. Which concept from this assignment did you find most interesting or surprising? Explain why.

The concept I found most interesting was Artificial Intelligence (AI) and Large Language Models (LLMs).

I was surprised by how AI systems can understand human language, answer questions and generate content such as text, images and code. I found it interesting because AI is becoming part of everyday life through tools such as ChatGPT, Microsoft Copilot and virtual assistant. Learning how these systems work gave me a better understanding of the technology that is shaping the future.

36. Which concept did you find most difficult to understand? What resources did you use to help you understand it better?

The concept I found most difficult to understand was Deep Learning and Neural Networks. At first, it was similar to how human brain processes information.

To improve my understanding, I used online learning resources such as IBM SkillsBuild, Microsoft Learn and education videos on YouTube. These resources explained the concepts using diagrams and real-world examples which made them easier to understand.

37. How do you see Data and AI affecting your future career or everyday life?

I believe that Data and AI will have a significant impact on both my future career and everyday life. In the workplace, data analytics can help organizations make better decisions by identifying trends and improving business processes.

AI tool can automate repetitive tasks, improve productivity and provide insights that would take much longer to discover manually.

In my everyday life, I already use AI through search engines, recommendation systems, navigation apps, virtual assistants. In the future, I expect AI to become even more useful in education, healthcare, transportation and personal productivity. Understanding data and AI will help me stay competitive in the job market and use technology more effectively.

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